Google Foundational Models & Vertex AI by Michaël Bettan

01. Core Definition & Architecture

Native Multimodality (Early Fusion) vs. Modular Perception Pipelines

Google's Gemini foundational models are engineered from the ground up around **native multimodality (early fusion)**. Instead of bolting separate vision or audio encoders onto a pretrained text-only backbone, all modalities — text, code, images, audio, video, and PDFs — are projected into a **unified, shared token space** during pretraining and processed by a single **Sparse Mixture-of-Experts (MoE)** transformer backbone.

Why bolted-on late-fusion cascades struggle on interactive multimodal tasks:

- **Inherent information loss:** Converting live audio or video into intermediate text strips paralinguistic cues (tone, hesitation, sarcasm, urgency) and fine-grained spatial-temporal alignment.
- **Compounded serial latency:** Chaining `ASR → Text LLM → TTS` creates sequential bottlenecks that break sub-second conversational voice and live video agents.
- **Cross-modal hallucination:** A text-dominant LLM frequently overrides visual or acoustic evidence across shallow adapter layers when signals conflict.

ARCHITECTURAL TRADE-OFF — NATIVE MULTIMODAL VS. MODULAR PIPELINE (PAA EXAM FOCUS)

Even with native multimodality, **modular pipelines** (`Chirp 3` / `Gemini 3.5 Transcribe` or `Document AI Layout Parser` → deterministic redaction/schema validation → text LLM) remain the right choice when compliance mandates **deterministic, auditable intermediate transcripts**, **PII redaction (`Sensitive Data Protection`) before reasoning**, custom domain phrase adaptation, or caching parsed text across many downstream passes. Choose **native multimodal** (`Gemini 3.8 Live` , `Gemini 4 Argon` GA agentic video understanding) when sub-second latency, prosody/visual nuance, or dynamic frame inspection is primary.

Evolution from PaLM 2 to Gemini Sparse MoE

- **PaLM 2 Baseline (Legacy, Retired):** Dense text-first transformer activating 100% of parameters on every token, requiring separate Vision or USM adapters.
- **Gemini Sparse Mixture-of-Experts (MoE):** Conditional learned routers dispatch each token to a specialized subset of expert feed-forward blocks — decoupling total model knowledge capacity from active per-token FLOPs, cost, and Time-to-First-Token (TTFT).

02. Active Ecosystem Tiers (October 2026 Lineup)

NAMING MILESTONE — GEMINI 4 PERIODIC-ELEMENT NAMING

Starting with **Gemini 4**, Google shifted frontier model naming to **periodic-table elements**, led by **Gemini 4 Argon (`gemini-4-argon` , GA October 13, 2026)** and **Gemini 4 Argon Cyber Permissive (`gemini-4-argon-cyber-permissive`)** (note: periodic-element naming starts at **Argon** — there is no "Gemini 4 Carbon" SKU).

Tier	Active flagship models (Oct 2026)	Architectural role & sweet spot
On-Device / Edge	**Gemini Nano** (Android AICore / Chrome Built-in AI), **Gemma 4 E2B / E4B**, **Gemma 3n**	Zero-cloud-cost, offline, privacy-preserving client inference under strict mobile/laptop RAM and battery envelopes
Flash-Lite (High-QPS)	**Gemini 3.5 Flash-Lite (`gemini-3.5-flash-lite`)**, `gemini-3.1-flash-lite`	Lowest cost & TTFT at massive scale: intent routing, inline guardrails, entity extraction, classification, bulk translation
Flash (Agentic Workhorse)	**Gemini 3.8 Flash (`gemini-3.8-flash`)**, `gemini-3.8-flash-cyber` , `gemini-3.7-flash` , `gemini-3.5-flash`	Default production agentic loop: long-horizon coding, high-velocity MCP/function calling, multimodal reasoning (default model in **CodeMender**)
Live, Speech & TTS	**Gemini 3.8 Live (`gemini-3.8-live`)**, `gemini-3.8-live-extended-thinking` , **3.8 Flash TTS**, **3.5 Transcribe**	Sub-second bidirectional voice/video agents with barge-in, background extended thinking during live speech, 130-language studio TTS, and low-latency diarized STT
Frontier Reasoning (Gemini 4)	**Gemini 4 Argon (`gemini-4-argon`)**, `gemini-4-argon-cyber-permissive` *(plus `gemini-3.1-pro` preview)*	Maximum System-2 reasoning for complex multi-step planning, repository-scale software engineering, legal/financial synthesis, math/science, computer use, and Fairwind-gated defensive cyber
Managed Autonomous Agents	**Antigravity Agent (`antigravity-preview-09-2026`)**, **Deep Research / Deep Research Max**, **CodeMender**	Task-level endpoints running multi-step browsing, coding, synthesis, or vulnerability patching inside managed harnesses/sandboxes

Legacy & retired lifecycle status: Gemini 2.5 (`2.5 Pro` , `2.5 Flash` , `2.5 Flash-Lite`) remains in legacy support for existing workloads (and managed SFT/preference tuning on Agent Platform), while new Gemini API projects default to **3.5 Flash-Lite**, **3.8 Flash**, and **4 Argon**. **Gemini 2.0 Flash / Flash-Lite**, `gemini-2.5-computer-use-preview-10-2025` , and standalone `Imagen 4` on Gemini API are shut down.

03. Model Garden & Serving Options

> **NAMING NOTE**
> Vertex AI Model Garden is now **Agent Platform Model Garden**, and Vertex AI MaaS is **Model as a Service (MaaS) on Agent Platform**.

Model Garden is the unified enterprise catalog of 200+ Google first-party, partner, and open-weight models with integrated evaluation, tuning, IAM, VPC-SC, and deployment controls.

The Three Model Families

Family	Representative models	Consumption model	Billing unit
First-party (Google)	Gemini 4 Argon, Gemini 3.8 Flash, 3.5 Flash-Lite, 3.8 Live, Nano Banana 2.1, Gemini Omni 1.1 Flash, Veo 3.1, Lyria 3.5, Chirp 3, Gemini Embedding 2	Fully managed serverless API only (cannot download weights)	Per token / media unit (PayGo) or reserved **GSUs** (Provisioned Throughput)
Partner (MaaS)	**Anthropic Claude**, **Mistral AI**, **DeepSeek** (e.g., `DeepSeek-V4.1-Flash` 1M context preview)	Managed serverless MaaS API inside Google Cloud governance	Per token (PayGo) or Provisioned Throughput where offered
Open-weight	**Gemma 4** (`E2B` , `E4B` , `26B A4B MoE` , `31B`), Gemma 3/3n, MedGemma, ShieldGemma 2, Llama, Qwen	**Either** curated serverless MaaS API **or** self-deployed on dedicated Agent Platform endpoints / **GKE**	MaaS: per token. Self-deployed: hourly VM + GPU/TPU compute **while deployed**

Partner Models on Agent Platform (Claude Example)

- **Google-managed Claude (`AnthropicVertex`):** Serverless inference inside Google Cloud's security perimeter with unified IAM, VPC-SC, Cloud Audit Logs, CMEK/data-residency controls, and Google Cloud billing.
- **Anthropic-managed Claude:** Marketplace-billed access to Anthropic-native platform features via a **separate** Anthropic organization (existing direct Anthropic org keys/ZDR configs do not transfer automatically).
- **Governance:** Enforce `constraints/vertexai.allowedPartnerModels` org policies to allowlist approved partner models. See the Anthropic & Claude guide.

Serving Open Models: MaaS vs. Dedicated Endpoint vs. GKE

Criterion	Serverless MaaS API	Self-deployed Agent Platform Endpoint	Self-deployed on GKE
Ops burden	Zero (Google autoscales accelerators)	Low–Medium (prebuilt vLLM / TGI / SGLang containers, autoscaling)	High (cluster, node pools, GPU drivers, serving stack)
Custom / LoRA weights	Base catalog models only	✅ Custom checkpoints, LoRA adapters, custom containers	✅ Full control + multi-LoRA dynamic serving
Traffic profile	Spiky, unpredictable, or prototyping	Steady, high utilization (lower TCO at sustained QPS)	Multi-model bin-packing across shared GPU/TPU pools
Billing trap	Pay strictly per token	**Billed continuously per node-hour even at 0 QPS** (configure idle undeploy)	Billed for provisioned node pools / reservations

04. Agentic Model Selection & Routing Framework

Golden rule: Pick the **smallest, fastest, cheapest model that passes your golden evaluation suite**, and compose models across roles rather than routing every agent turn through a frontier model.

Multi-Model Agentic Routing Pattern

1. **Front-door router & inline guardrail (`Gemini 3.5 Flash-Lite` or `ShieldGemma 2`):** Classifies intent, extracts structured slots, screens policy, and answers trivial FAQs in `<200 ms` at minimal cost.
2. **Primary execution & tool loop (`Gemini 3.8 Flash`):** Executes multi-turn MCP tool calls, code generation, and RAG synthesis at high token velocity.
3. **Frontier escalation / planner (`Gemini 4 Argon`):** Invoked selectively for ambiguous multi-step decomposition, deep repository refactoring, legal/financial synthesis, or conflict resolution when confidence is low.
4. **Vertex AI Model Optimizer:** Managed dynamic router that automatically balances quality vs. cost/latency across Gemini tiers per prompt based on your target preference profile.

LLM vs. SLM & Open-Weight vs. Proprietary

- **LLM (Gemini 4 Argon / 3.8 Flash) vs. SLM (Gemma 4 / Gemini Nano / 3.5 Flash-Lite):** Use LLMs for open-ended planning, massive 1M multimodal context, and zero-shot generalization; use SLMs (especially fine-tuned LoRA SLMs or on-device LiteRT-LM / Android AICore) for high-QPS narrow schemas, strict edge latency, or offline client privacy.
- **Proprietary vs. Open-Weight:** Proprietary APIs deliver peak frontier capability (`Gemini 4 Argon`) with managed safety and zero infra ops, but enforce vendor retirement schedules (e.g., `gemini-4-argon` has a published earliest retirement horizon of Oct 2027+). Open weights (`Gemma 4`) give permanent weight ownership, air-gapped deployment, and custom fine-tuning, **but you own serving, patching, and external guardrails** (**Model Armor** / **ShieldGemma 2**).
- **Exam trap:** *"Open-weight"* does **not** force self-hosting — consume open models via **MaaS** first unless you need custom weights or dedicated VPC isolation.

Customization Ladder: Prompting → Grounding/RAG → Tuning

Rung	Solves for	Key architectural rule
1. Prompt & Context Engineering	Task framing, structured JSON schemas, dynamic few-shot (**Example Store**)	Always establish an eval-measured prompt baseline first (and use **Agent Optimizer** for automated prompt tuning)
2. Grounding / RAG	**Missing, private, or volatile knowledge** + verifiable citations	Never fine-tune to inject changing facts or ACL-protected documents; use **Agent Search**, **RAG Engine**, or Search Grounding
3. Supervised Fine-Tuning (SFT)	**Consistent behavior**: strict domain syntax, terse classification labels, latency/token reduction by replacing huge few-shot prompts	Supported via LoRA adapter tuning on eligible Gemini 2.5/3.x and Gemma models (**not supported on `gemini-4-argon`**)
4. Preference Tuning (RLHF/DPO)	Subtle tone, helpfulness, or stylistic trade-offs hard to specify as exact targets	Run after SFT using `(prompt, preferred, dispreferred)` pairs

Consumption Tiers: PayGo, Flex, Batch & Provisioned Throughput

Tier	Target workload	Behavior & exam nuances
Standard PayGo	Variable interactive traffic	Shared capacity; subject to `429` under regional spikes (use `global` endpoint where allowed)
Priority PayGo	Latency-critical bursts without term lock-in	Per-request priority header on shared pools (supported on `gemini-4-argon`, `3.8-flash`, `nano-banana-2.1`); can absorb PT overflow
Flex PayGo	Synchronous but latency-tolerant jobs (evals, bulk tagging)	Discounted token rate in exchange for variable latency and higher preemption
Batch Inference	Massive asynchronous offline pipelines	Deepest token discount (typically ~50% off Standard) with asynchronous turnaround
Provisioned Throughput (PT)	Mission-critical production needing guaranteed QPS/TPM & p95 SLA	Reserve **GSUs** for a fixed non-cancelable term (1 wk, 1 mo, 3 mo, 1 yr). **Spills over to PayGo by default**; send `X-Vertex-AI-LLM-Request-Type: dedicated` to cap spend strictly (`429` on overage) or `shared` to bypass PT

05. Gemini 4 Argon, Gemini 3.x & Context Mechanics

Deep Dive: Gemini 4 Argon (`gemini-4-argon`) Mechanics & Constraints

Released GA on **October 13, 2026**, **Gemini 4 Argon** introduces architectural behaviors that differ sharply from Gemini 2.5 and 3.x — **high-yield exam territory**:

- **Unified 1M Token Window (`1,048,576` tokens):** Unlike earlier models with separate input and output pools, `gemini-4-argon` shares a **single unified `1,048,576`-token budget** across **input prompt + internal thinking tokens + function calls + generated output**.
- **Long Decodes up to 1M Tokens:** Default single synchronous response cap is **`65,536` tokens** (`32,768` on non-streaming HTTP), **extendable all the way to `1,048,576` tokens (minus input)** for whole-codebase generation or exhaustive legal/scientific synthesis using: 1. `continuation_token` loops on `GenerateContent` (`v1beta1`), **or** 2. Server-side `background=true` execution via the **`Interactions API`** (`client.interactions.create` with `previous_interaction_id`).
- **Strict `thinking_level` Validation (`LOW` | `MEDIUM` | `HIGH`):** Defaults to **`MEDIUM`**. Passing legacy integer **`thinking_budget`** OR passing **`thinking_level="MINIMAL"`** on `gemini-4-argon` fails immediately with HTTP **`400 INVALID_ARGUMENT`**! (Note: `MINIMAL` is also rejected on `gemini-3.8-flash` and `gemini-3.7-flash`, but is supported on `gemini-3.6-flash` and `gemini-3.5-flash*`; integer `thinking_budget` belongs to Gemini 2.5, where `2.5 Pro` cannot be set to `0`.)
- **Sampling Parameters Ignored:** Custom `temperature`, `top_p`, and `top_k` values are **ignored** on `gemini-4-argon` (reasoning calibration is managed internally).
- **Implicit Context Caching Only:** `gemini-4-argon` uses **automatic durable implicit caching** (minimum **4,096 tokens**, zero storage fee, up to 95% input discount on Gemini API). **Explicit `cachedContents` creation, Managed Tuning, Live API WebSockets, and direct RAG Engine generator binding are NOT supported on `gemini-4-argon`.**
- **Native Agentic Capabilities:** Supports **GA Agentic Video Understanding** (actively zooming/re-sampling video regions during reasoning), native **Computer Use** (GUI screen action loops), Code Execution, MCP/Function Calling, URL Context, and Search/Maps/Agent Search/Enterprise Web Grounding.
- **Gemini 4 Argon Cyber Permissive (`gemini-4-argon-cyber-permissive`):** Gated behind Google DeepMind's **Fairwind Program** allowlist (`global` and `us` endpoints) for authorized defensive cybersecurity teams performing vulnerability discovery, reverse engineering, exploit verification, and automated patching with **CodeMender** (`cm find / verify / fix --model gemini-4-argon`). Defaults to `HIGH` thinking and **disables Computer Use** as a containment control.

`GenerateContent` vs. Stateful `Interactions API` (Preview)

Capability	`GenerateContent` API	`Interactions API` (`client.interactions.create`)
State model	Client-managed stateless request (must resend conversation history + **Thought Signatures** each turn)	Optional **server-side state** via `previous_interaction_id` — preserves multi-turn context and reasoning state automatically
Long-horizon async	Synchronous / SSE (`continuation_token` needed past 65K output tokens)	Native `background=true` mode for multi-minute/multi-hour autonomous decodes and **Managed Agents** (`Antigravity` , `Deep Research`)
Best fit	Low-latency custom orchestrators, fine-grained context compaction	Long-decode synthesis, stateful multi-turn agents, and hosted Managed Agents

Thought Summaries, Thought Signatures & Context Caching Across Generations

- **Thought Summaries & Thought Signatures:** Set `include_thoughts=True` to receive human-readable reasoning summaries for UI observability; pass opaque cryptographic **Thought Signatures** back in multi-turn `GenerateContent` tool loops so the model retains its latent reasoning chain without re-deriving steps.
- **Implicit vs. Explicit Caching Comparison:**
- **Implicit Caching:** Zero config, zero hourly storage charge; triggers automatically on shared prompt prefixes (`⩾4,096` tokens on Gemini 4 Argon). Keep static instructions and tool definitions at the top of the prompt!
- **Explicit Caching (`cachedContents`):** Supported on Gemini 2.5 and eligible Gemini 3.x base models (not fine-tuned models, **not Gemini 4 Argon**); bills hourly storage for a deterministic TTL cache handle.
- **ZDR & Multi-Tenant Safety:** Caches are isolated per project and honor VPC-SC, but disable caching when strict **Zero Data Retention (ZDR)** applies, and never share explicit cache handles across tenants with distinct data entitlements.

Multimodal Tokenization Mechanics

- **Text:** 256,000-token **SentencePiece** vocabulary.
- **Images:** Dynamic tiling into **768×768** pixel tiles (**258 tokens per tile**), plus configurable media resolution controls on Gemini 3.x/4 to trade detail vs. token budget.
- **Video:** Sampled visual frames + audio track (**~263 tokens/sec at 1 FPS** standard resolution; compressible to **~66–70 tokens/sec** in low-resolution mode for multi-hour recordings), with **GA Agentic Vision Understanding** dynamically inspecting high-res regions on demand.
- **Audio:** Compact **32 tokens per second** (~1,920 tokens/minute) interleaved with timestamp markers (`MM:SS`) for exact temporal grounding.

06. Generative Media Models (Images, Video & Music)

Nano Banana 2.1 / Pro & Imagen 3 / 4 (Visual Synthesis & Editing)

- **Nano Banana Family — Native Multimodal Image Generation (Recommended Default):**
- **Nano Banana 2.1 (`gemini-nano-banana-2.1` , GA + Priority PayGo):** Flagship native Gemini image generator and conversational editor, superseding standalone `Imagen 4` on the Gemini API.
- **Nano Banana 2 (`gemini-3.1-flash-image`) & Nano Banana 2 Lite (`gemini-3.1-flash-lite-image`):** High-speed, cost-efficient native image generation and mask-free multi-turn editing.
- **Nano Banana Pro (`gemini-3-pro-image`):** Studio-grade **4K** generation with **131,072-token** context, search-grounded factual infographics/typography, and multi-reference subject consistency (up to **14 reference images**: e.g., 5 character + 6 object + 3 style references).
- **Imagen 3 (`imagen-3.0-generate-002`) & Imagen 4 (`imagen-4.0-generate-001` / `ultra` / `fast` on Agent Platform):** Specialized latent-diffusion text-to-image endpoints (480-token prompt window, up to 2K) with **SynthID** watermarking and **C2PA Content Credentials** (note: standalone Imagen 4 is retired on the developer Gemini API in favor of Nano Banana 2.1, while remaining available on enterprise Agent Platform).

Gemini Omni 1.1 Flash & Veo 3.1 (Video + Native Audio Generation)

- **Gemini Omni 1.1 Flash (`gemini-omni-1.1-flash`):** Fast native multimodal video generation, instruction-driven video editing, keyframe interpolation, and clip extension with synchronized native audio (replaces legacy Veo 3.1 preview endpoints on the Gemini API).
- **Veo 3.1 (`veo-3.1-generate-001` , `veo-3.1-fast-generate-001` , `veo-3.1-lite-generate-001` — GA on Agent Platform): Joint latent diffusion** compressing video pixels and audio waveforms into a shared spatio-temporal latent space — producing 4K 16:9 / 9:16 cinema-grade clips with frame-accurate lip-sync, Foley sound effects, and object permanence across camera pans (succeeding silent **Veo 2**).

Lyria 3.5, Lyria 3 & RealTime (Music Generation)

- **Temporal Latent Diffusion:** Compresses 44.1–48 kHz stereo PCM audio into compact latents and denoises structured musical waveforms without autoregressive long-horizon drift.
- **Active Lineup: Lyria 3.5 (`lyria-3.5`)** (flagship full-length songs with multi-section structural coherence and multilingual vocals), **Lyria 3 Pro** (184s compositions), **Lyria 3 Clip** (30s rapid hooks), **Lyria RealTime (`lyria-realtime-exp`)** (interactive live-steered music streaming), and **Lyria 2 (`lyria-002`)** — all protected by **SynthID-Audio**, C2PA, and artist vocal-likeness/recitation filters.

07. Live Voice Agents, Speech (3.8 TTS / 3.5 Transcribe / Chirp 3) & Embeddings

Gemini Multimodal Live API (`gemini-3.8-live`)

- **Stateful Bidirectional WebRTC / WebSockets:** Persistent low-latency streaming sessions ingesting continuous **16-bit PCM audio**, live camera/screen frames, and text while emitting native **24 kHz audio**.
- **Active Live Models:**
- **Gemini 3.8 Live (`gemini-3.8-live`):** Default sub-second voice & multimodal agent model with server-side VAD **barge-in**, affective prosody adaptation, and spoken function calling (**71.5%+ ComplexFuncBench Audio**).

- **Gemini 3.8 Live Extended Thinking (`gemini-3.8-live-extended-thinking`):** Performs asynchronous high-depth background reasoning while maintaining natural conversational responsiveness during live voice sessions.
- *(Note: `gemini-4-argon` does not support the Live API — pair `gemini-3.8-live` at the voice front-door with `gemini-4-argon` sub-agents for deep backend tasks!)*

Dedicated Speech Synthesis, Transcription & Live Translation

- **Gemini 3.8 Flash TTS (`gemini-3.8-flash-tts`) & 3.8 Flash-Lite TTS (`gemini-3.8-flash-lite-tts`):** Studio-fidelity controllable text-to-speech across **130 languages** (101 on Flash-Lite TTS) with natural-language style/voice design and low-latency streaming.
- **Gemini 3.5 Transcribe (`gemini-3.5-transcribe` / `gemini-3.5-transcribe-live`):** Dedicated low-latency speech-to-text with speaker diarization, per-utterance multilingual switching, and domain vocabulary biasing.
- **Gemini 3.5 Live Translate (`gemini-3.5-live-translate-preview`):** Real-time speech-to-speech translation preserving speaker prosody across **70+ languages**.
- **Chirp 3 (`chirp_3` in Cloud Speech-to-Text V2):** Built on Google's **2B+ parameter Universal Speech Model (USM)** (12M+ hours, 100+ languages) for enterprise compliance pipelines needing `Streaming Recognize` / `BatchRecognize` , **Model Adaptation** (up to 1,000 custom domain phrases), `denoise_audio=True` , word-level timestamps, and deterministic redaction before LLM ingestion.

Gemini Embedding 2 & `gemini-embedding-001`

- **Gemini Embedding 2 (`gemini-embedding-2` / `gemini-embedding-2-preview`):** Google's **first natively multimodal Gemini embedding model**, mapping **text, code, images, audio, video, and PDFs** directly into a single unified vector space — enabling cross-modal retrieval (e.g., matching a smartphone photo of a faulty valve to both a PDF schematic and a repair-video timestamp) without captioning pipelines.
- **Matryoshka Representation Learning (MRL):** Both `gemini-embedding-2` and text-focused `gemini-embedding-001` (up to 3,072 dimensions) order semantic signal from index `0` onward, letting architects truncate `output_dimensionality` (e.g., `1536` , `768` , `512` , `256`) with automatic L2 normalization to cut Vector Search / AlloyDB RAM by 2–12× at minimal recall loss.

08. Gemma 4 & Open-Weight Edge Architecture

The **Gemma** family brings Google DeepMind architectures to open weights for Model Garden MaaS, GKE/dedicated endpoints, air-gapped on-prem servers, and **Google AI Edge** (`LiteRT-LM` / MediaPipe).

Gemma 4, Gemma 3n & Gemma 3 Evolution

- **Gemma 4 (`E2B` , `E4B` , `26B A4B MoE` , `31B`):** Latest generation open-weight family combining edge-optimized **Per-Layer Embeddings (PLE)** on `E2B` / `E4B` (processing **text, image, video, and native audio** in a 2B–4B effective RAM footprint), a high-efficiency **26B Sparse MoE activating only 4B parameters per token (`26B A4B`)** for low-latency self-hosted agentic tool loops, and a dense **31B** reasoning checkpoint.
- **Gemma 3 (1B, 4B, 12B, 27B) & Gemma 3n (E2B, E4B):** Introduced **128K context** (140+ languages), **SigLIP** 400M vision encoding (256 soft tokens per **896×896 Pan & Scan** crop), **5:1 local-to-global interleaved attention** (5 sliding-window 1,024-token layers per 1 global layer to shrink KV-cache RAM), and **MatFormer** nested sub-networks.
- **T5Gemma & RecurrentGemma (Griffin):** Encoder-decoder open checkpoints (`T5Gemma`) for sequence-to-sequence tasks, alongside **Griffin (`RG-LRU`)** constant-memory linear recurrence for ultra-long edge streaming.

Specialized Gemma Ecosystem Variants

- **ShieldGemma 2 (4B) & ShieldGemma (2B, 9B, 27B):** Open text + image safety classifiers scoring prompts, completions, and visual inputs against harm taxonomies (ideal as fast local guardrails around self-hosted open models).
- **MedGemma (4B, 27B) & MedSigLIP (800M):** Clinical text, EHR summarization, and radiology/pathology/ophthalmology multimodal comprehension (the open-weight successor alongside Gemini to legacy **MedLM**).
- **TxGemma (2B, 9B, 27B):** Therapeutic small-molecule, protein, and clinical-trial prediction models.
- **PaliGemma 2 (3B, 10B, 28B), CodeGemma (2B, 7B) & FunctionGemma:** Specialized VLM fine-tuning (bounding boxes/segmentation), IDE Fill-in-the-Middle (FIM), and compact tool-calling adapters.

09. Safety, Provenance & Frontier Governance

SynthID Imperceptible Watermarking + C2PA

Metadata manifests (**C2PA Content Credentials**) provide rich cryptographic provenance chains when intact, but are stripped by screenshots or re-encoding. **SynthID** embeds imperceptible signatures directly into generated signals for defense-in-depth:

- **SynthID-Text:** Seeded cryptographic **g-function tournament sampling** over token logits without degrading perplexity or latency.
- **SynthID-Audio:** Psychoacoustic spectral watermarking surviving MP3 compression, pitch/tempo shifts, and the **"analog hole"** (speaker playback re-recorded via microphone).
- **SynthID-Image & Video:** Spatial-temporal latent perturbation resilient to cropping, color grading, and lossy transcoding.

DeepMind Frontier Safety Framework (FSF) & Enterprise Guardrails

- **Critical Capability Levels (CCLs):** Empirical pre-deployment evaluations across **CBRN uplift**, **Autonomous Cyber-Offense** (governing why `gemini-4-argon-cyber-permissive` is gated behind the **Fairwind Program** with Computer Use disabled), **Autonomous ML R&D / self-improvement**, and **Deceptive alignment / oversight circumvention**.
- **Coupled Mitigations: Security Mitigations** (weight exfiltration hardening) + **Deployment Mitigations** (safety tuning, runtime classifiers, allowlisted access).
- **Enterprise Runtime Defense:** Combine Gemini `SafetySetting` harm thresholds with **Vertex AI Model Armor** (prompt injection, jailbreak, malicious URL, and SDP PII screening) and **ShieldGemma 2**.

10. Scientific, Embodied & Agentic Discovery Systems

Multi-Agent Scientific & Algorithmic Discovery (Agentic Frontier)

- **AI Co-Scientist:** Multi-agent scientific research system built on Gemini where specialized agents (**Generate → Reflect → Rank → Evolve → Proximity → Meta-review**) compete in an Elo-rated self-play tournament over asynchronous tasks to synthesize novel, experimentally validated biomedical and scientific hypotheses (e.g., drug repurposing and epigenetic targets).
- **AlphaEvolve:** Evolutionary coding agent pairing Gemini's creative code synthesis with **automated deterministic evaluators** in an evolutionary loop — discovering provably correct, novel algorithms for matrix multiplication (surpassing Strassen's 1969 4×4 complex bound), open mathematical problems, and Google data-center Borg scheduling / TPU Verilog kernel optimizations.
- **CodeMender:** Autonomous defensive security agent powered by **Gemini 3.8 Flash** (default) and **Gemini 4 Argon** (`cm find`, `cm verify`, `cm fix`) that discovers vulnerabilities, proves root causes with sanitizers/fuzzers, and generates regression-tested patches.

Structural Biology, Mathematics, Robotics & World Models

- **AlphaFold 3:** Replaces Evoformer with a 48-block **Pairformer**, **atomic-level tokenization** for ligands/ions, and a **3D Conditional Diffusion Module** predicting joint complexes across proteins, DNA, RNA, small molecules, and covalent modifications.
- **AlphaProteo & AlphaMissense:** *De novo* high-affinity protein binder generation (3×–300× affinity uplift) and pathogenicity classification across 71M human missense variants.
- **AlphaGeometry 2:** Neuro-symbolic IMO gold-standard prover pairing a **Gemini** intuitive auxiliary-construction generator ("System 1") with the **DDAR** symbolic deduction verifier ("System 2").
- **Gemini Robotics ER 2 (`gemini-robotics-er-2-preview`):** Next-generation **Embodied Reasoning** Vision-Language-Action model succeeding `Gemini Robotics ER 1.6` (retired) and RT-2 — performing 3D spatial pointing, grasp/trajectory planning, task progress estimation, and physical safety reasoning for dexterous robots.
- **Genie 2 / 3 & SIMA 2:** Interactive physics-consistent 3D world generation from single prompts (`Genie`) and generalist language-following 3D virtual agents (`SIMA 2`).
- **GraphCast / GenCast, AlphaChip, AlphaQubit, AlphaTensor & AlphaDev:** GNN/diffusion global weather ensembles, RL TPU floorplanning, surface-code quantum error decoding, and RL-discovered matrix/LLVM sorting algorithms.